

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Traversal Challenge

Course Code:	CSA0305	Course Title:	Data Structures
CO Assessed:	CO3 – Precision (BL3, BL5)	Assessment:	Assessment Tool 2 – Traversal Challenge
Weightage:	30% of CIA	Total Marks:	100
CO3 Statement:	<i>Solve problems for optimized data storage and retrieval using Binary Search Trees, AVL Trees, B-Trees, Red-Black Trees, and Splay Trees.</i>		
Student Name:	S.Roshni	Reg. No.:	192525185
Section / Batch:	AI&ML	Date:	03/08/2026

Code of Conduct

I S.Roshni(192525185)_____ (Name / Reg No)
certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: S.Roshni_____

Instructions

- This assessment tool contains 5 Traversal Challenge questions, Total: 100 Marks.
- Students can use any online/offline Traversal Challenge. Demonstrate tree.
- When a student answer using simulator, in evaluation the answer should have captured screenshots after every major operation
- Submit the report in PDF format with properly labelled screenshots as proof of completion.

Section / Topic		Focus	Q Nos	Marks
Tree Traversals		Traversal types, In order, Post order and Pre order	5	100
GRAND TOTAL:			100 Marks	

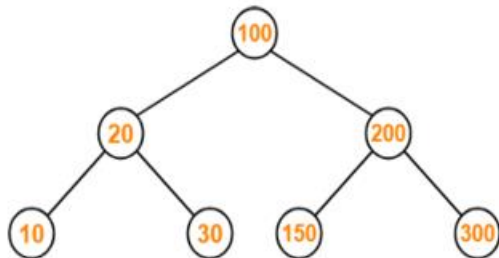
Traversal Challenge

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Conceptual Understanding	20	Demonstrates thorough understanding of all core Data Structures concepts; answers accurately reflect deep knowledge	Shows good understanding with minor conceptual gaps	Basic understanding; several misconceptions present	Limited understanding; major concepts misunderstood
Traversal Accuracy	20	Correctly completes all tree traversals (Preorder, Inorder, Postorder, Level Order) without errors.	Completes most traversals correctly with one minor error.	Correctly performs only some traversals with multiple errors.	Unable to perform most traversals correctly.
Selection of Appropriate Traversal	30	Correctly selects the appropriate traversal for every given problem scenario.	Selects the correct traversal for most scenarios.	Selects appropriate traversal for only a few scenarios.	Frequently selects incorrect traversal methods.
Completion & Time Efficiency	20	Completes all challenges within the allotted time while maintaining accuracy.	Completes most challenges within the time limit.	Completes only part of the challenge or exceeds the time limit slightly.	Unable to complete the challenge within the allotted time.
Evidence Submission	10	Uploads complete screenshots showing successful completion, score, and student details.	Uploads screenshots with minor missing information.	Uploads incomplete or unclear screenshots.	No valid evidence submitted.

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Question No:1 Write the traversal sequences for following trees



a. Preorder

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Tree Traversals

Input
Input is level-by-level order.

100,20,200,10,30,150,300

Enter numbers separated by commas. Use null for empty nodes.

Traversal Order

Pre In Post

Generate Default

Visualize

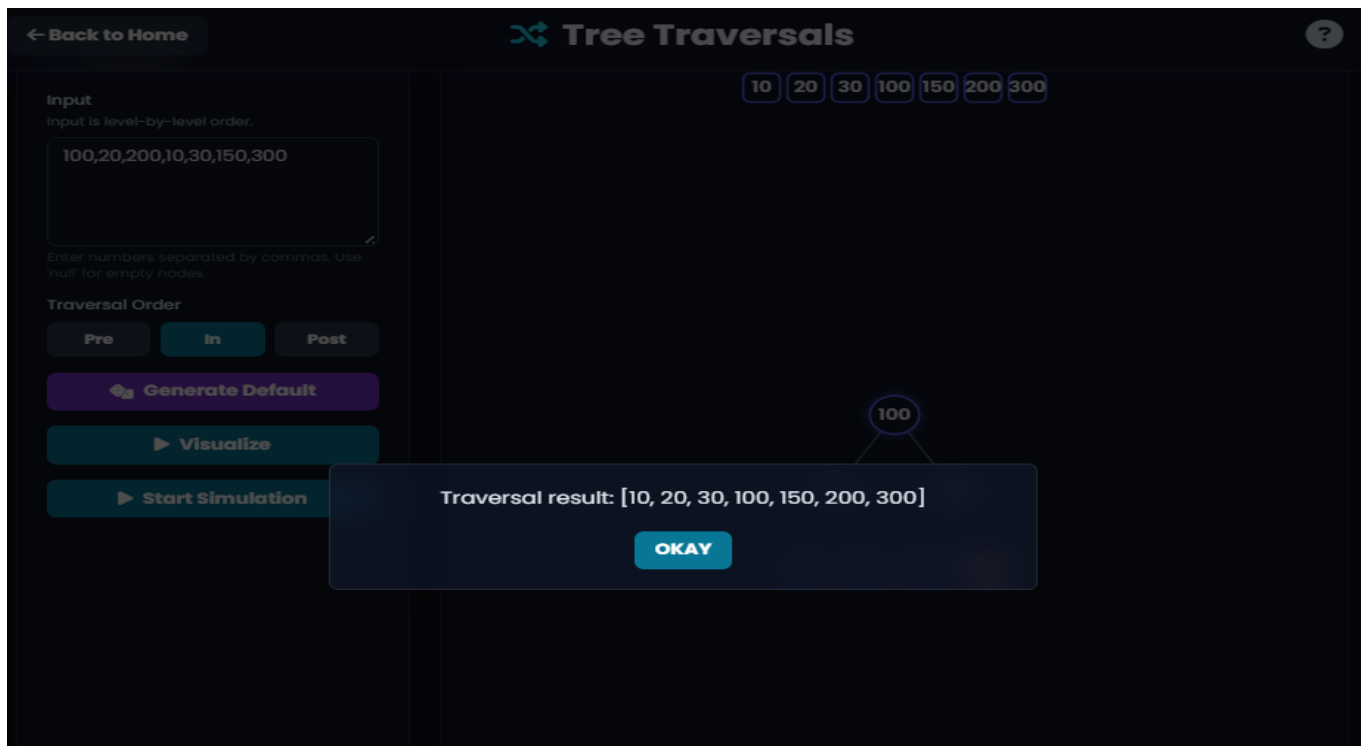
Start Simulation

10 30 20 150 300 200 100

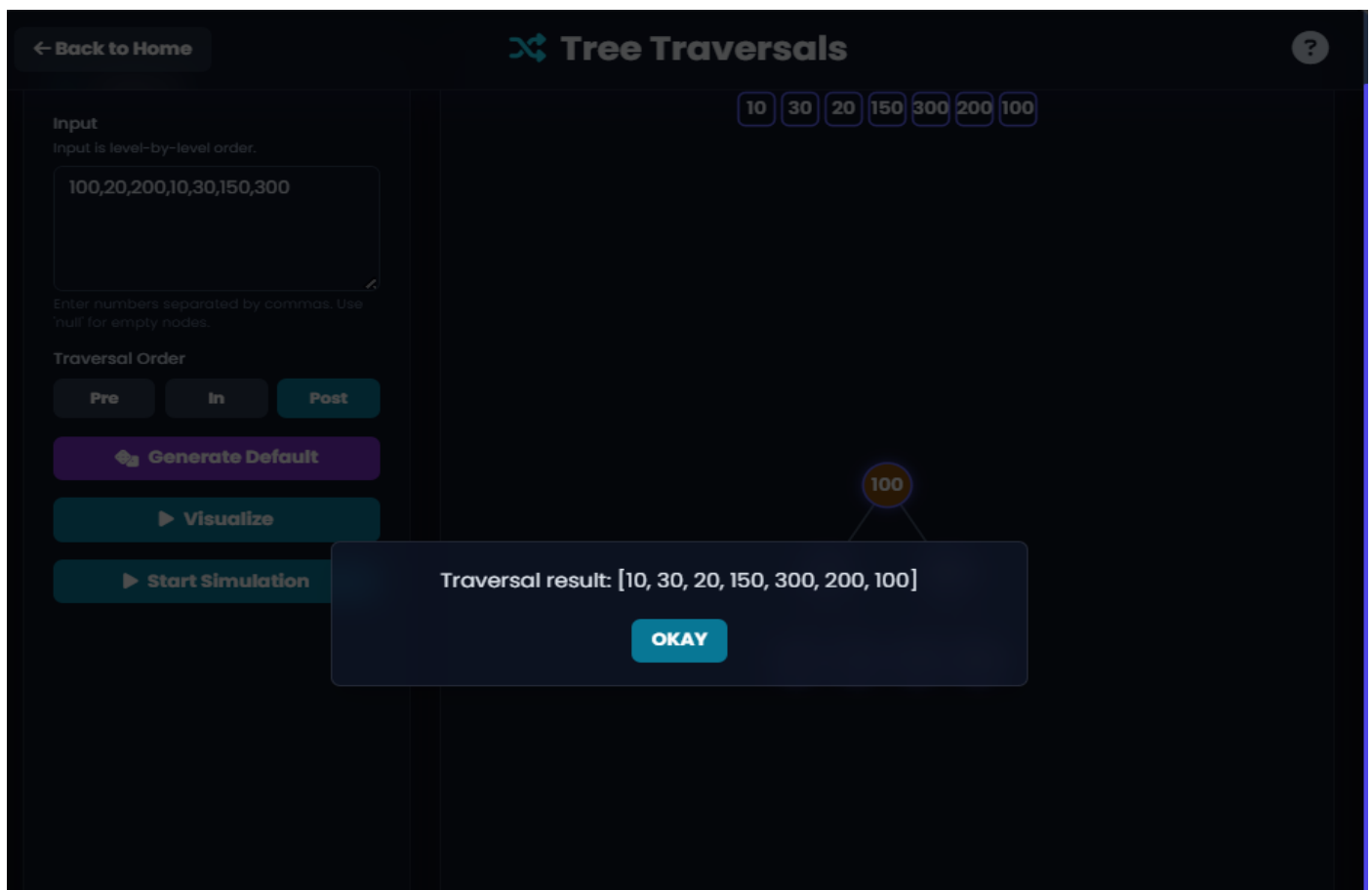
Traversal result: [10, 30, 20, 150, 300, 200, 100]

OKAY

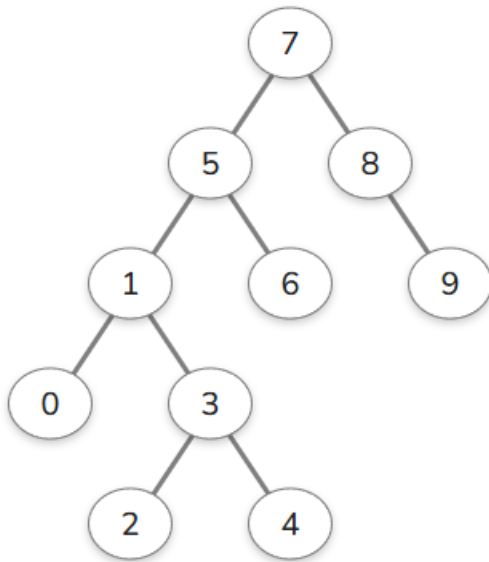
b. Inorder



c. Postorder



Question No: 2 Assume the digits 7, 5, 1, 8, 3, 6, 0, 9, 4, 2 are added in that sequence into a binary search tree that is initially empty. The binary search tree follows the standard natural number ordering. What is the resulting tree's traversal sequence?



Preorder:

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Tree Traversals

Input
Input is level-by-level order.

7,5,8,1,6,9,0,3,2,4

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre In Post

Generate Default

Visualize

Start Simulation

7 5 1 3 2 6 4 8 9 0

7
5 8
1 6 9
0 3 2 4

Traversal result: [7, 5, 1, 3, 2, 6, 4, 8, 9, 0]

OKAY

Inorder

:

The screenshot shows the 'Tree Traversals' application interface. On the left, the 'Input' field contains '7,5,8,1,6,9,0,3,2,4'. The 'Traversal Order' section has 'In' selected. A modal dialog displays the 'Traversal result: [3, 1, 2, 5, 4, 6, 7, 9, 8, 0]'. The tree diagram shows a root node 7 with left child 5 and right child 8. Node 5 has left child 3 and right child 2. Node 8 has left child 4 and right child 0. The input sequence is shown at the top: 3, 1, 2, 5, 4, 6, 7, 9, 8, 0.

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Tree Traversals

Input

Input is level-by-level order.

7,5,8,1,6,9,0,3,2,4

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre In Post

Generate Default

Visualize

Start Simulation

Traversal result: [3, 1, 2, 5, 4, 6, 7, 9, 8, 0]

OKAY

3 1 2 5 4 6 7 9 8 0

7

5 8

3 2 4

0

Postorder:

The screenshot shows the 'Tree Traversals' application interface. On the left, the 'Input' field contains '7,5,8,1,6,9,0,3,2,4'. The 'Traversal Order' section has 'Post' selected. A modal dialog displays the 'Traversal result: [3, 2, 1, 4, 6, 5, 9, 0, 8, 7]'. The tree diagram shows a root node 7 with left child 5 and right child 8. Node 5 has left child 3 and right child 2. Node 8 has left child 4 and right child 0. The input sequence is shown at the top: 3, 2, 1, 4, 6, 5, 9, 0, 8, 7.

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Tree Traversals

Input

Input is level-by-level order.

7,5,8,1,6,9,0,3,2,4

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre In Post

Generate Default

Visualize

Start Simulation

Traversal result: [3, 2, 1, 4, 6, 5, 9, 0, 8, 7]

OKAY

3 2 1 4 6 5 9 0 8 7

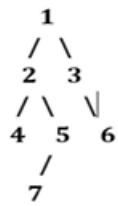
7

5 8

3 2 4

0

Question No: 3 Write the Inorder, Preorder & Postorder traversal sequences for following trees



Preorder:

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Tree Traversals

Input

Input is level-by-level order.

1,2,3,4,5,6,7

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre In Post

Generate Default

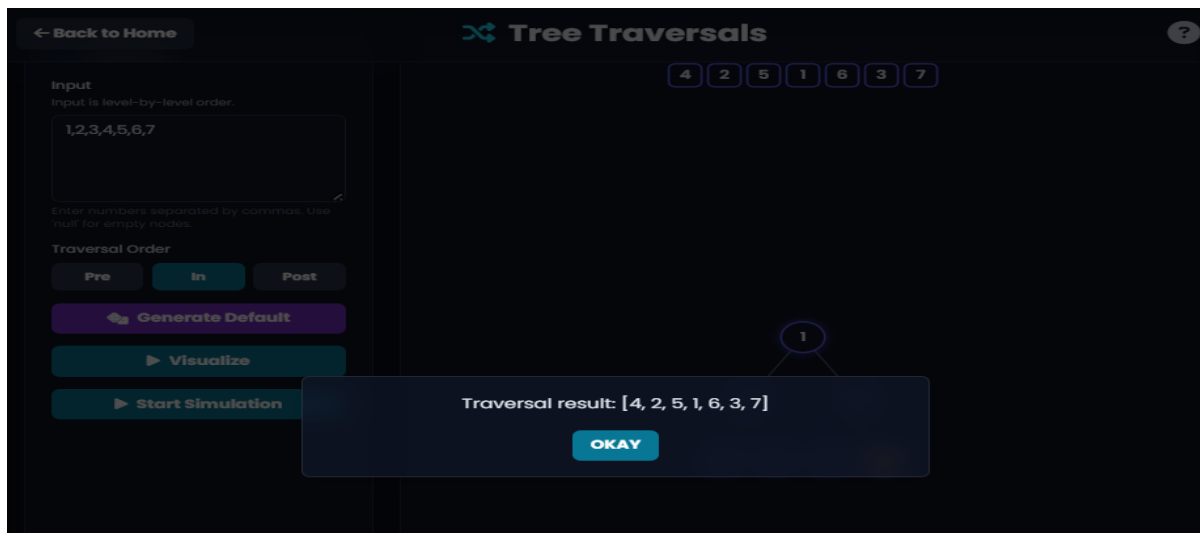
Visualize

Start Simulation

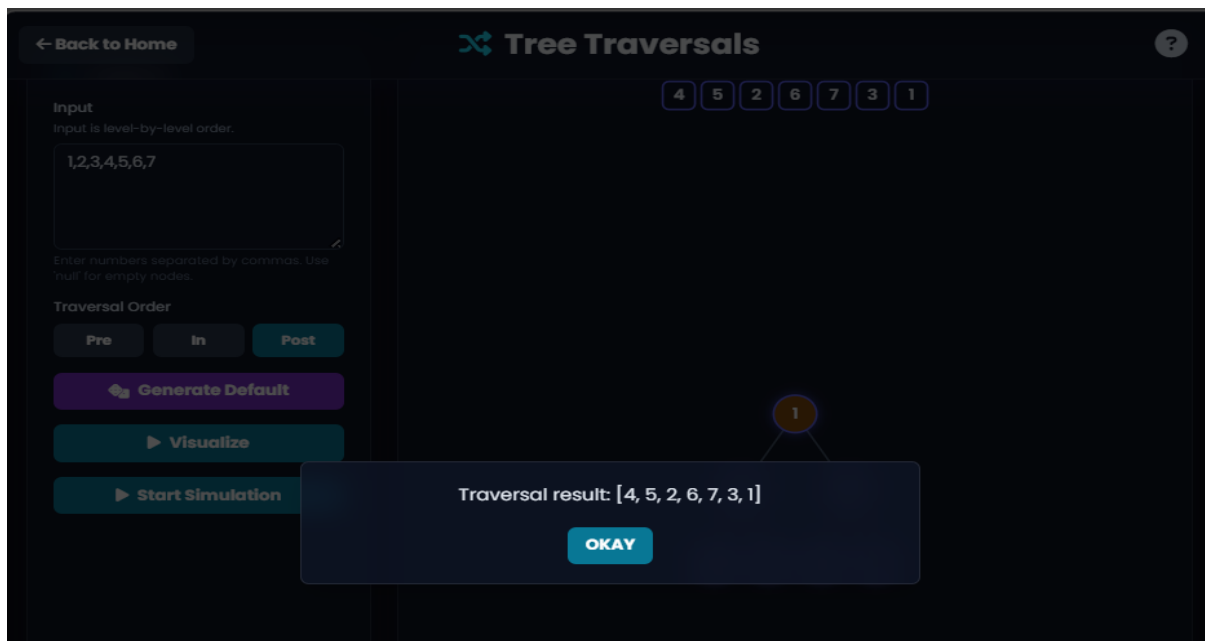
Traversal result: [1, 2, 4, 5, 3, 6, 7]

OKAY

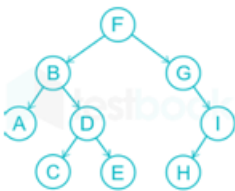
Inorder:



Postorder:



Question No: 4 What is the sequence of nodes when applying in-order traversal on the binary tree given below?



Inorder:

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Tree Traversals
?

Input
 Input is level-by-level order.

F,B,G,A,D,I,C,E,H

Enter letters separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

Post

Generate Default

Visualize

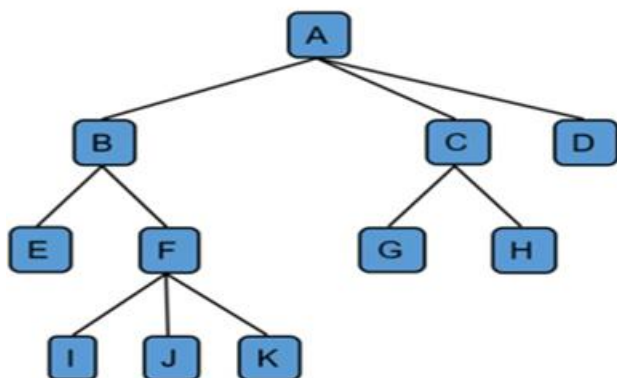
Start Simulation

F B G A D I C E H

Inorder Traversal Result:
[A, B, C, D, E, F, G, H, I]

OKAY

Question No: 5



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Tree Traversals
?

Input
 Input is level-by-level order.

A,B,C,D,E,F,G,H,I,J,K

Enter letters separated by commas. Use null or # for empty nodes.

Traversal Order

Pre

In

Post

Generate Default

Visualize

Start Simulation

A B C D E F G H I J K

Traversal Result (Inorder):
[E, B, I, F, J, K, A, G, C, H, D]

OKAY

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Tree Traversals

?

A

B

C

D

E

F

G

H

I

J

K

Input
Input is level-by-level order.

A,B,C,D,E,F,G,H,I,J,K

Enter letters separated by commas.
Use null or # for empty nodes.

Traversal Order

Pre

In

Post

Generate Default

Visualize

Start Simulation

```
graph TD; A((A)) --- B((B)); A --- C((C)); A --- D((D)); B --- E((E)); B --- F((F)); C --- G((G)); C --- H((H)); F --- I((I)); F --- J((J)); F --- K((K));
```

Preorder Traversal Result:

[A,B,E,F,I,J,K,C,G,H,D]

OKAY

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Tree Traversals

?

A

B

C

D

E

F

G

H

I

J

K

Input
Input is level-by-level order.

A,B,C,D,E,F,G,H,I,J,K

Enter letters separated by commas.
Use null or # for empty nodes.

Traversal Order

Pre

In

Post

Generate Default

Visualize

Start Simulation

```
graph TD; A((A)) --- B((B)); A --- C((C)); A --- D((D)); B --- E((E)); B --- F((F)); C --- G((G)); C --- H((H)); F --- I((I)); F --- J((J)); F --- K((K));
```

Postorder Traversal Result:

[E,I,J,K,F,B,G,H,C,D,A]

OKAY